Hands-on Learning Platforms Directory

Programming • Cybersecurity • Artificial Intelligence • Student benefits

A practical comparison guide to 20 platforms and student offers. Use it as a catalogue to explore at your own pace—not as a checklist you must complete. Access and pricing can change, so check each official page before committing.

A. Programming practice and problem-solving

Platform	Purpose	Specialties / topics	How you learn	Level / difficulty	Cost / access	What you may want from it / official page
LeetCode	Algorithm and data-structure practice; especially common interview-style problems.	Arrays, strings, hash maps, trees, graphs, dynamic programming, SQL, complexity and optimization.	Solve individual problems; topic filters, study plans and timed contests.	Easy → Hard; beginner-friendly entry, but hard problems become demanding.	Freemium: many problems and contests are free; Premium unlocks selected content/features.	Best main practice site if you want to keep problem-solving sharp in short sessions. **Official page:** https://leetcode.com/problemset/
Codeforces	Competitive programming platform built around timed contests and a public rating system.	Algorithms, mathematics, implementation speed, greedy methods, graphs, dynamic programming and contest strategy.	Rated contests, problem archive, rating progression and editorials/community solutions.	Beginner → elite; the contest environment can feel steep at first.	Free to use for core contests and problems.	Best when you want competition, measurable progress and pressure-tested problem solving. **Official page:** https://codeforces.com/
AtCoder	Japanese competitive-programming platform with structured contest series.	Algorithmic reasoning, implementation, mathematics and progressively harder contest problems.	AtCoder Beginner Contest (ABC), Regular Contest (ARC), Grand Contest (AGC), practice archive and ratings.	Beginner → elite; ABC is the usual entry point.	Free for core contests and problems.	A structured alternative to Codeforces; start with ABC contests. **Official page:** https://atcoder.jp/
Codewars	Practice platform where small programming exercises are called 'kata'.	Language fluency, idiomatic code, fundamentals, algorithms, refactoring and problem decomposition.	Short challenges, community solutions and Kyu/Dan ranks; supports many languages.	Beginner → advanced; 8 kyu is the easiest rank, moving toward 1 kyu and Dan.	Freemium; core kata practice is available free, with optional paid features.	Good for short, satisfying practice and learning how other programmers express a solution. **Official page:** https://www.codewars.com/
Exercism	Language-learning and fluency platform built around curated exercises and feedback.	85 programming languages, language idioms, readable code, debugging and core concepts.	Guided language tracks, in-browser editor, automated analysis and optional human mentoring.	Beginner → advanced; choose a language track that matches your level.	100% free, according to Exercism.	Excellent if you want to maintain or improve a particular language rather than compete. **Official page:** https://exercism.org/
Project Euler	A collection of mathematical and computational programming problems.	Number theory, primes, combinatorics, probability, sequences, mathematical reasoning and efficient algorithms.	Independent problems that often require deriving a mathematical insight and then coding it.	Intermediate → very challenging; some early problems are approachable.	Free registration and problem access.	Especially rewarding if you enjoy mathematics and want programming to reinforce mathematical thinking. **Official page:** https://projecteuler.net/
HackerRank	Structured coding practice and technical skill assessments.	Algorithms, data structures, SQL, Python, Java, C++, REST APIs and interview-oriented skills.	Topic-based exercises, practice tracks, timed assessments and skill certifications.	Beginner → advanced, depending on track.	Core practice is free; HackerRank says its skill certifications are free. Some employer/enterprise features differ.	Useful for structured topic practice, SQL, and checking skills with timed assessments. **Official page:** https://www.hackerrank.com/

B. Cybersecurity and ethical-hacking labs

Platform	Purpose	Specialties / topics	How you learn	Level / difficulty	Cost / access	What you may want from it / official page
PortSwigger Web Security Academy	A free, hands-on curriculum for learning web application security in safe labs.	SQL injection, XSS, CSRF, authentication, access control, SSRF, API testing, request smuggling, business logic and more.	Explanations followed by interactive browser-based vulnerability labs; difficulty labels from Apprentice to Expert.	Beginner → expert; individual topics vary.	100% free for Academy learning materials and labs; Burp Suite Community Edition can be used for practice.	Top recommendation for learning how to find and understand web application vulnerabilities. **Official page:** https://portswigger.net/web-security/
TryHackMe	Guided, gamified cybersecurity learning with practical rooms and learning paths.	Linux, networking, web security, penetration testing, defensive security, forensics and fundamentals.	Step-by-step rooms, browser-based or VPN-connected labs, guided paths and exercises.	Beginner → advanced; strong starting point for newcomers.	Freemium: free rooms and limited lab/tool access; full paths and greater access require a subscription.	Good when you want a guided route rather than being dropped into an unexplained machine. **Official page:** https://tryhackme.com/
Hack The Box	Hands-on penetration-testing practice using vulnerable machines and security challenges; also has a separate Academy.	Linux/Windows, network services, web exploitation, privilege escalation, Active Directory and pentesting workflows.	Standalone machines/challenges, guided Starting Point content, Academy modules and larger lab environments.	Beginner entry exists, but much of the platform suits intermediate → advanced learners.	Freemium: some entry content is free; many machines, Academy modules and features are paid.	A strong next step after basic networking, Linux and web-security foundations. **Official page:** https://www.hackthebox.com/
OverTheWire	Free security wargames that teach practical command-line and systems concepts.	Linux shell, SSH, file permissions, processes, networking basics, cryptography and other security topics.	Level-by-level games; Bandit is designed for absolute beginners and unlocks the next level as you progress.	Beginner → advanced depending on the wargame; start with Bandit.	Free.	A lightweight way to build Linux and terminal confidence in short sessions. **Official page:** https://overthewire.org/wargames/
picoCTF	Educational Capture The Flag challenges developed by Carnegie Mellon University.	Web exploitation, cryptography, forensics, reverse engineering, binary exploitation and general security puzzles.	Challenge-based CTF environment with hints, flags and learning resources.	Beginner → advanced; designed to be approachable for students.	Free practice platform.	Great student-friendly variety; useful for exploring different cybersecurity specialisms. **Official page:** https://picoctf.org/
Root-Me	A broad challenge platform for practical information-security skills.	Web server/client, network, cryptography, programming, forensics, reverse engineering and system security.	Large library of standalone challenges with points, progression and community activity.	Beginner → advanced; challenge difficulty varies considerably.	Free challenges are available; optional paid features/services may exist—check current account options.	Good for browsing many security categories and finding a niche that interests you. **Official page:** https://www.root-me.org/

C. Artificial intelligence and machine learning — learn by doing

Platform	Purpose	Specialties / topics	How you learn	Level / difficulty	Cost / access	What you may want from it / official page
Kaggle Learn	Short, practical courses and notebooks for data science and machine learning.	Python, pandas, data visualization, SQL, introductory/intermediate ML, deep learning, computer vision, NLP and model evaluation.	Browser-based lessons, coding exercises, notebooks, datasets and competitions.	Beginner → intermediate for Learn courses; competitions can be advanced.	Courses are provided at no cost; Kaggle also provides a free learning and competition ecosystem.	Best practical starting point for coding small ML models and working with real datasets. **Official page:** https://www.kaggle.com/learn/
Google Machine Learning Crash Course	A practical introduction to the concepts and mathematics behind machine learning.	Regression, classification, data preparation, overfitting, neural networks, embeddings, LLM basics, production ML and fairness.	Short lessons, videos, interactive visualizations, quizzes and browser-based programming exercises.	Beginner to intermediate; Python and algebra/statistics familiarity help.	Free.	Best when you want to understand why ML methods work, not only how to call a library. **Official page:** https://developers.google.com/machine-learning/crash-course
TensorFlow Playground	A visual browser experiment for understanding how neural networks learn.	Neural-network layers, neurons, activation functions, learning rate, training behavior and decision boundaries.	Change network settings and data features, then watch training and classification boundaries update.	Beginner; little setup or coding required.	Free and open-source.	A quick visual intuition-builder; not a full course or a substitute for coding models. **Official page:** https://playground.tensorflow.org/
fast.ai — Practical Deep Learning for Coders	A practical course that teaches deep learning by building working models.	PyTorch, computer vision, NLP, tabular data, model training, evaluation and deployment concepts.	Lessons paired with notebooks and projects; emphasizes working models early, then explaining the theory.	Intermediate; programming experience and comfort with Python are recommended.	Course materials are free; compute resources may have separate costs depending on where you run notebooks.	Good once you know Python and want to build and train real deep-learning models. **Official page:** https://course.fast.ai/
Hugging Face Learn	A collection of courses around open-source machine learning tools and modern AI models.	Transformers, LLMs, NLP, datasets, tokenizers, diffusion, reinforcement learning and AI agents.	Guided courses, code examples, notebooks and projects using the Hugging Face ecosystem.	Beginner → advanced depending on course; some tracks assume Python/ML basics.	Course materials are generally free; hosted compute, APIs or third-party services may have usage costs.	Best for exploring how modern language and generative models are used, adapted and built with open tools. **Official page:** https://huggingface.co/learn

D. Student Developer Pack extras mentioned earlier (not standalone learning platforms)

Platform	Purpose	Specialties / topics	How you learn	Level / difficulty	Cost / access	What you may want from it / official page
Educative (GitHub Student Developer Pack offer)	Interactive, text-based courses with browser coding playgrounds and hands-on labs.	Web development, Python, Java, machine learning and other software topics.	Guided interactive lessons and in-browser coding; course catalogue varies.	Beginner → advanced depending on course.	GitHub currently lists 6 months of access to 70+ practical courses for verified students, plus a stated 30% subscription discount; confirm current terms before redeeming.	Potentially useful if you want structured lessons and practice in one place; the student offer is time-limited. **Official page:** https://education.github.com/pack
Astra Security (GitHub Student Developer Pack offer)	Website security product rather than a general cybersecurity course.	Website firewall, malware scanning and managed bug-bounty-related services.	Security tooling for protecting a website; not primarily a lesson-and-lab curriculum.	Not a learning-level platform; requires a website/project context.	GitHub currently lists 6 months of website firewall and malware-scanner access for eligible students; verify redemption conditions.	Included because it was mentioned earlier, but don't confuse this with a hands-on ethical-hacking training site. **Official page:** https://education.github.com/pack

A sensible way to explore the list

1. Keep LeetCode as your main programming habit; one or two problems in a spare session is enough.

2. For cybersecurity, try OverTheWire Bandit for terminal fundamentals and PortSwigger Academy for web vulnerabilities.

3. For AI, start with Kaggle Learn or Google’s Machine Learning Crash Course. Use TensorFlow Playground as a visual companion.

4. Treat the other platforms as a catalogue, not a checklist. Pick one platform at a time based on what genuinely interests you.

5. Only test security techniques in authorized labs or systems you own or have explicit permission to assess.

Verification note: A fully free course, a limited free tier, and a time-limited student offer are different types of access. GitHub Student Developer Pack benefits are listed on GitHub Education’s official Pack page; verify current eligibility and redemption terms.